

Decoding Policies and Sinusoidal Positions

From a next-token distribution to generated text — and from token indices to position vectors.

PART 1
Temperature, Top- k , Top- p

PART 2
Beam Search

PART 3
Sinusoidal Positional Embedding

Lecture Overview

We leave the forward pass fixed and ask two separate questions about what surrounds it.

THREE PARTS OF THE LECTURE

SAMPLING POLICIES

Temperature, top- k , top- p reshape the next-token distribution

Each policy keeps the model fixed and changes only how a token is drawn from $p_{\theta}(w_{t+1} \mid w_{\leq t})$.

BEAM SEARCH

A small set of candidate sequences, scored by log-probability

Beam search keeps the B best prefixes at each step and returns the highest-scoring complete sequence.

SINUSOIDAL POSITIONS

A fixed function that assigns each index a position vector

The sinusoidal construction makes relative position recoverable as a linear map on the position vectors.

We start where last lecture ended — the next-token distribution — and ask how to choose from it.

The Decoding Question

The model emits a distribution; a decoding policy chooses the token.

THE FORWARD PASS PRODUCES A DISTRIBUTION

$$p_{\theta}(w_{t+1} \mid w_{\leq t}) = \text{softmax}(u_t)$$

TOKEN	PROBABILITY
dark	0.57
red	0.26
grey	0.12
cold	0.05

Context: "The sky turned ..."

GREEDY DECODING

Take the arg-max at every step.

Deterministic, repetitive, blind to any lower-probability continuation that leads to a better full sequence.

PURE RANDOM SAMPLING

Draw $w_{t+1} \sim p_{\theta}$.

Faithful to the distribution, but the long tail of low-probability tokens accumulates into incoherent text.

EVERYTHING WE BUILD TODAY

Reshape or truncate p_{θ} before sampling, or widen the search beyond one token.

Greedy and pure sampling are two endpoints; the practical policies live between them.

Temperature Sampling

A single scalar divides the logits and reshapes the softmax.

DEFINITION

$$p_T(w_i) = \frac{\exp(z_i/T)}{\sum_j \exp(z_j/T)}$$

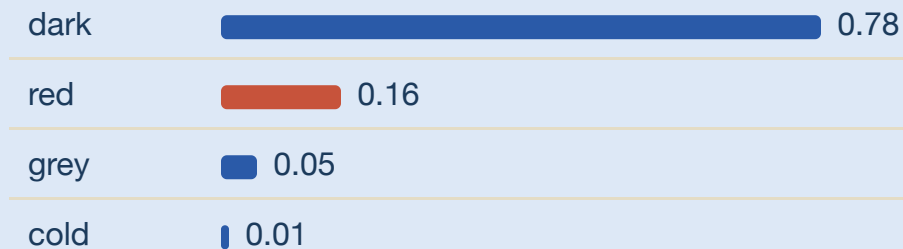
$T = 1$ — the model's own distribution.

$T \rightarrow 0$ — all mass concentrates on the arg-max; equivalent to greedy decoding.

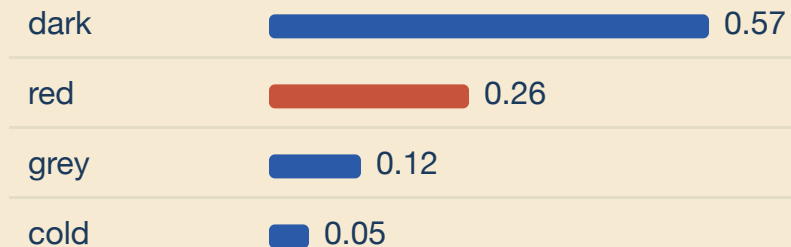
$T \rightarrow \infty$ — the distribution flattens toward uniform over the vocabulary.

RESHAPING THE SAME LOGITS

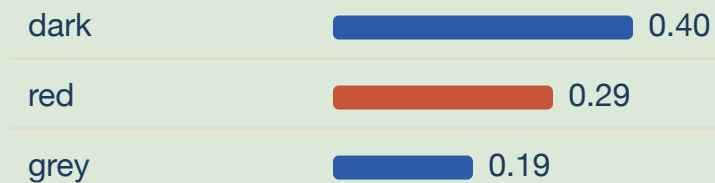
COOL · $T = 0.5$



NEUTRAL · $T = 1$



WARM · $T = 2$



Temperature changes how peaked the distribution is, but never removes any token.

Top-*k* Sampling

Keep the *k* most probable tokens, renormalize, and sample.

PROCEDURE

$$\mathcal{V}_k = \underset{v}{\text{top-}k} \, p_{\theta}(v \mid w_{\leq t})$$

$$\tilde{p}(w) = \begin{cases} p_{\theta}(w) / Z_k & w \in \mathcal{V}_k \\ 0 & \text{otherwise} \end{cases}$$

$Z_k = \sum_{v \in \mathcal{V}_k} p_{\theta}(v)$ is the renormalizing constant over the kept set.

The long tail is **removed**, not just downweighted.

EXAMPLE · *K* = 3

TOKEN	P_{θ}	KEPT?	P
dark	0.57	yes	0.60
red	0.26	yes	0.27
grey	0.12	yes	0.13
cold	0.05	no	0
...tail...	...	no	0

Small *k* — safe, conservative, risks repetition.

Large *k* — closer to pure sampling, long tail returns.

Weakness — a fixed *k* ignores how peaked the distribution already is.

*Top-*k* truncates by count: always exactly *k* candidates, regardless of their mass.*

Top- p (Nucleus) Sampling

Keep the smallest set of tokens whose total probability exceeds p .

PROCEDURE

$$\mathcal{V}_p = \arg \min_{\mathcal{V} \subseteq V} |\mathcal{V}| \quad \text{s.t.} \quad \sum_{v \in \mathcal{V}} p_{\theta}(v) \geq p$$

$$\tilde{p}(w) = \begin{cases} p_{\theta}(w) / Z_p & w \in \mathcal{V}_p \\ 0 & \text{otherwise} \end{cases}$$

Sort tokens by probability, take them from the top until the cumulative mass crosses the threshold p , then renormalize.

The size of $\mathcal{V}_p$ **adapts** to how peaked the distribution is at that step.

EXAMPLE · $P = 0.9$

TOKEN	P_{θ}	CUM.	KEPT?
dark	0.57	0.57	yes
red	0.26	0.83	yes
grey	0.12	0.95	yes — crosses p
gold	0.05	1.00	no

CONTEXT	$ \mathcal{V}_P $
Peaked — one token dominates	often 1–3
Flat — many plausible continuations	can be 30+

Top- p truncates by mass: the candidate set grows and shrinks with the model’s confidence.

Comparing and Combining the Policies

In practice, temperature is composed with a top- k or top- p truncation.

TEMPERATURE
What it changes
The shape of the whole distribution.
Vocabulary kept
All $|V|$ tokens remain in the support.
Risk
Low T repeats; high T drifts into gibberish.

TOP- K
What it changes
The support — exactly k tokens remain.
Vocabulary kept
Fixed count, regardless of peakiness.
Risk
Too wide at peaked steps, too narrow at flat ones.

TOP- P
What it changes
The support — adaptive to the step.
Vocabulary kept
As few tokens as needed to cover mass p .
Risk
Needs a sensible p ; still samples randomly within the nucleus.

TYPICAL PIPELINE (APPLIED IN ORDER)

$$\text{logits} \xrightarrow{/T} \text{softmax} \xrightarrow{\text{top-}p} \tilde{p} \xrightarrow{\text{sample}} w_{t+1}$$

Temperature sets how confident the model acts; top- p cuts the long tail; sampling draws the token.

These policies compose: reshape with temperature, truncate the tail, then sample.

From Token-at-a-Time to Sequence Search

Sampling and greedy both commit after one step; beam search carries alternatives forward.

WHY ONE-STEP CHOICES ARE MYOPIC

The arg-max token at step t may force a *worse* whole sequence than a lower-ranked token that unlocks a strong continuation at step $t + 1$.

$$\log p_{\theta}(w_{1:T}) = \sum_{t=1}^T \log p_{\theta}(w_t \mid w_{<t})$$

Scoring a full sequence sums log-probabilities along its path.

Exact search is intractable. The tree branches by $|V|$ at every step; $|V|^T$ paths cannot be enumerated for any realistic T .

GREEDY

Width 1 — one path only.

BEAM SEARCH

Width B — the B best prefixes at every step.

EXHAUSTIVE

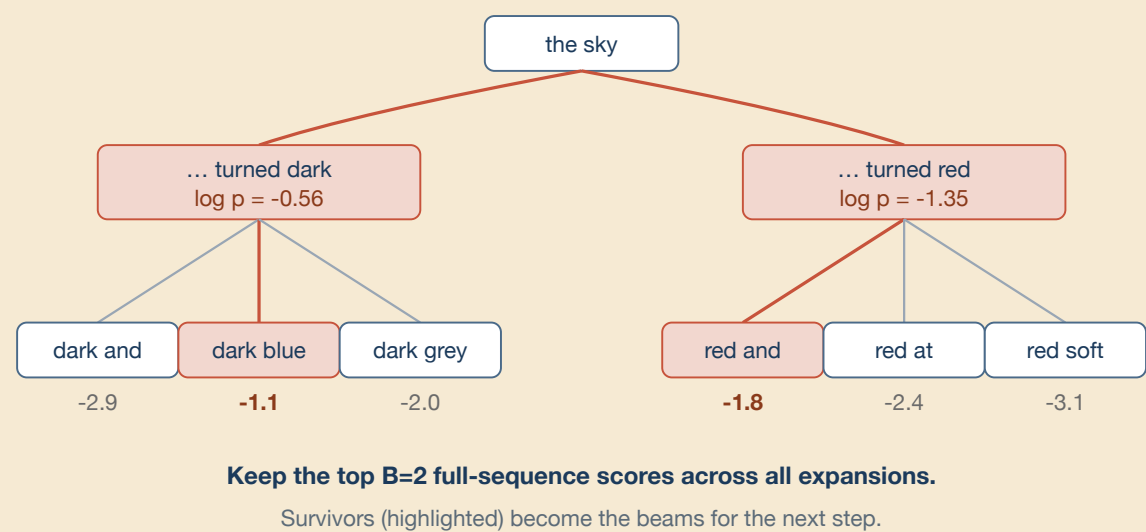
Width $|V|^T$ — not feasible.

Beam search is a bounded compromise: a few paths, scored by total log-probability.

Beam Search Mechanics

At each step, expand every surviving prefix by every token, then keep the top B .

ONE BEAM STEP ($B = 2$)



ALGORITHM

1 • Initialize

Start with one beam holding the prefix (or a start token). Score 0.

2 • Expand

For each of the B beams, compute $p_{\theta}(w \mid \text{prefix})$ for every w in the vocabulary.

3 • Score

Each expansion gets score $s + \log p_{\theta}(w \mid \text{prefix})$.

4 • Prune

Across the $B \cdot |V|$ expansions, keep the top B by score.

5 • Stop

Finish a beam when it emits the end-of-sequence token; return the best completed beam.

Beam search is best-first local search over a growing frontier of B partial sequences.

Beam Width and Length Normalization

Widening the beam does not monotonically improve generation.

BEAM WIDTH B

$B = 1$ — greedy decoding.

Larger B — explores more alternatives, higher-scoring sequences found.

Too-large B — in open-ended generation, favors short and generic outputs. For machine translation, $B \in [4, 10]$ is typical.

Cost — each decoding step runs the model on B prefixes in parallel.

LENGTH BIAS

Longer sequences accumulate more negative log-probabilities, so the raw score

$$\log p_{\theta}(w_{1:T}) = \sum_{t=1}^T \log p_{\theta}(w_t \mid w_{<t})$$

systematically prefers **shorter** sequences.

LENGTH-NORMALIZED SCORE

$$\text{score}(w_{1:T}) = \frac{1}{T^{\alpha}} \sum_{t=1}^T \log p_{\theta}(w_t \mid w_{<t})$$

$\alpha \in [0.6, 1.0]$ dampens the penalty on long sequences. $\alpha = 0$ recovers the raw score.

Beam search balances search width against decoding cost and a built-in bias toward short outputs.

Which Decoding Policy, When?

The right choice is set by whether the task has one correct output or many plausible ones.

LOW-ENTROPY TASKS

Translation · summarization · code generation · constrained question answering

A single near-correct output exists. Favor **beam search** (or greedy at $B = 1$) with length normalization.

Sampling introduces unnecessary variance; truncation parameters are not needed.

HIGH-ENTROPY TASKS

Open-ended generation · dialogue · creative writing · brainstorming

Many good continuations exist. Favor **sampling with top- p** and a chosen temperature.

Beam search collapses toward safe, repetitive text in this regime.

COMMON DEFAULTS IN PRODUCTION DECODERS

TEMPERATURE	Values near 0 for deterministic tasks; 0.7–1.0 for conversational models.
TOP- P	Usually in $[0.9, 0.95]$ for open-ended generation.
TOP- K	Values near $k = 40\text{--}50$ when used; often combined with or replaced by top- p .
BEAM WIDTH	$B = 4\text{--}10$ for translation; rarely used for open-ended generation.

The decoding policy is a knob on the task, not on the model.

Why the Transformer Needs Position Information

Self-attention is permutation-equivariant; order has to be injected explicitly.

WITHOUT POSITIONS

$$\text{Attn}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V$$

Permute the rows of the input and the output rows permute identically.

The model cannot distinguish “*dog bites man*” from “*man bites dog*” from the token identities alone.

ADD A POSITION VECTOR TO EACH TOKEN

$$x_i^{(0)} = e_i + s(i)$$

e_i is the learned token embedding at index i .

$s(i)$ encodes the absolute position i in the sequence.

Learned absolute — a lookup table of position vectors trained with the model.

Sinusoidal — a fixed function with no learned parameters; the subject of the next slides.

Position has to be added to the token row before block 0; the question is what function to use.

Sinusoidal Positional Embedding

A pair of sine and cosine for each dimension pair, with geometrically spaced frequencies.

DEFINITION

$$\text{PE}(\textit{pos}, 2i) = \sin\left(\frac{\textit{pos}}{10000^{2i/d}}\right)$$

$$\text{PE}(\textit{pos}, 2i + 1) = \cos\left(\frac{\textit{pos}}{10000^{2i/d}}\right)$$

$\textit{pos}$ — the position index (0-indexed).

i — the dimension-pair index, $0 \leq i < d/2$.

d — the model dimension (matches the token embedding).

No learned parameters. Every row is a fixed function of its index.

READING THE CONSTRUCTION

Each dimension pair $(2i, 2i + 1)$ carries one angular frequency

$$\omega_i = \frac{1}{10000^{2i/d}}$$

The frequencies form a **geometric sequence** from 1 (low i) to $\frac{1}{10000}$ (high i).

Low i · fast sinusoids — change quickly across positions, resolve fine distinctions.

High i · slow sinusoids — change slowly, carry coarse absolute location.

Together: a *frequency ladder* from fine to coarse along the depth axis of the position vector.

Each position becomes a d -vector of sines and cosines at geometrically spaced rates.

The Frequency Ladder

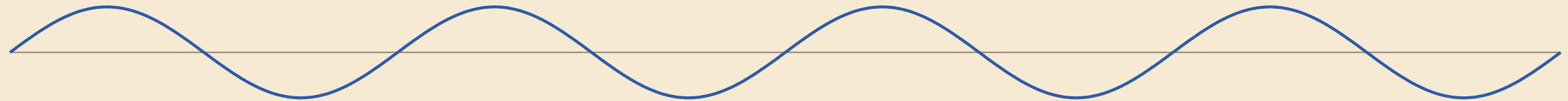
Fast sinusoids in the first dimensions, slow sinusoids deeper in the vector.

ROWS: DIMENSION PAIRS • COLUMNS: POSITION INDEX POS

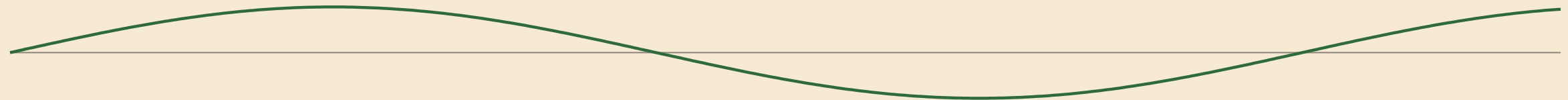
$I = 0 \cdot \Omega = 1$



$I \text{ SMALL} \cdot \Omega \approx 1/10$



$I \text{ MEDIUM} \cdot \Omega \approx 1/100$



$I \text{ LARGE} \cdot \Omega \approx 1/10000$



pos = 0

pos $\rightarrow \infty$

Each dimension-pair carries one angular frequency; higher-index dimensions oscillate ever more slowly.

The position vector is a spectrum of oscillations, from fine to coarse along i .

Why Sinusoids? Relative Position as a Linear Map

For any fixed offset k , $\text{PE}(pos + k)$ is a linear function of $\text{PE}(pos)$.

ANGLE ADDITION INSIDE EACH DIMENSION PAIR

Fix a dimension pair i with frequency ω_i . The two coordinates at position pos are

$$\begin{bmatrix} \sin(\omega_i pos) \\ \cos(\omega_i pos) \end{bmatrix}$$

Shift the position by k and apply $\sin(a+b)$ and $\cos(a+b)$:

$$\begin{bmatrix} \sin(\omega_i(pos+k)) \\ \cos(\omega_i(pos+k)) \end{bmatrix} = \begin{bmatrix} \cos(\omega_i k) & \sin(\omega_i k) \\ -\sin(\omega_i k) & \cos(\omega_i k) \end{bmatrix} \begin{bmatrix} \sin(\omega_i pos) \\ \cos(\omega_i pos) \end{bmatrix}$$

A pure **rotation by angle $\omega_i k$** — depending only on the offset k , never on pos .

WHAT THIS GIVES THE TRANSFORMER

Stacking the $d/2$ rotation blocks into one matrix M_k ,

$$\text{PE}(pos + k) = M_k \text{PE}(pos)$$

The map from position pos to position $pos + k$ is a **fixed linear transformation** determined entirely by k .

Consequence — the dot-product attention score

$$\langle W_Q \text{PE}(pos), W_K \text{PE}(pos + k) \rangle$$

is a function the model can learn to read as a *relative* position signal: differences of positions become learnable through linear operations.

Extrapolation — the formula is defined for every pos , including positions longer than any seen during training.

Sinusoids turn position differences into rotations the attention mechanism can read linearly.

Sinusoidal vs Learned Absolute Positions

Two ways to populate $s(i)$ — one fixed, one trained.

LEARNED ABSOLUTE

$$s(i) = E_{\text{pos}}[i]$$

Parameters — one d -vector for every index up to L_{max} , trained with the model.

Capacity — the model is free to choose any geometry across positions.

Limitation — rows beyond L_{max} do not exist; the model cannot process longer sequences without new training.

Used by — GPT-2, BERT, many encoder-decoder NMT systems.

SINUSOIDAL (FIXED)

$$s(i) = \text{PE}(i)$$

Parameters — none; a fixed function of the index.

Capacity — the geometry is set by the frequency ladder, not learned.

Extrapolation — the formula is defined at every integer index, including positions longer than those seen in training.

Used by — the original Transformer; the direct ancestor of today's relative schemes (RoPE, ALiBi).

WHERE THIS SITS IN THE FORWARD PASS

$$X^{(0)} = E_{\text{tok}}[w_{1:N}] + s(1:N)$$

Positions enter once, at the bottom of the stack, and are carried through the residual stream.

Learned positions give capacity; sinusoidal positions give structure and extrapolation for free.

Lecture Summary

Two independent choices that sit on either side of the Transformer stack.

DECODING POLICY — ABOVE THE MODEL

- 1 **Temperature** reshapes the softmax; low T sharpens, high T flattens.
- 2 **Top- k** truncates to a fixed count of candidates.
- 3 **Top- p** truncates by cumulative mass and adapts to the step.
- 4 **Beam search** keeps B prefixes and scores by total log-probability, with length normalization to counter the short-sequence bias.

POSITIONAL ENCODING — BELOW THE MODEL

- 1 Self-attention is permutation-equivariant: position must be injected into $X^{(0)}$.
- 2 **Sinusoidal** embeddings assign each dimension pair a geometrically spaced frequency.
- 3 A shift of k positions acts as a **fixed rotation** on each dimension pair — relative position is a linear function of absolute position.
- 4 Learned absolute positions give capacity but cap at $L_{\max}$; sinusoids extrapolate for free.

Next lecture: relative-position schemes (RoPE, ALiBi) that fold position directly into attention.